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# MSQF 533: Financial Risk Management

## Unit IV – Management of Financial Risk

### Meaning of interest rate risk

**Interest rate risk is the possibility of losing money because market interest rates change.**

Interest rates affect the cost of borrowing, the value of bonds, investment returns, and the profits of banks and companies.

For example, suppose you purchase a bond that pays a fixed interest rate of 8% per year. Later, newly issued bonds begin offering 10%.

Investors may prefer the new bonds because they offer higher returns. Therefore, the market price of your existing 8% bond may fall.

This reduction in the market value of your investment is an example of interest rate risk.

### Simple illustration

**8% -** Your bond's interest rate

**10%-** New market interest rate

When market interest rates rise, the price of an existing fixed-rate bond generally falls.

### Causes of interest rate risk

- Changes in the Reserve Bank of India's monetary policy.
- Inflation and inflation expectations.
- Changes in demand for and supply of money.
- Changes in economic growth.
- Government borrowing requirements.
- Changes in global interest rates.

### Types of interest rate risk

**Repricing risk:** Interest rates on assets and liabilities change at different times.

*Example:* A bank gives a five-year loan at a fixed rate but accepts deposits whose interest rates are revised every year. If deposit rates rise, the bank's funding costs may increase while its loan income remains unchanged.

**Basis risk:** Two interest rates that usually move together begin to move differently.

*Example:* A company has a loan linked to one benchmark rate and a hedge linked to another. If the two benchmarks move differently, the hedge may not fully protect the company.

**Yield curve risk:** Changes in the shape or slope of the yield curve affect investments differently.

*Example:* Short-term interest rates may rise while long-term rates remain unchanged.

**Embedded-option risk:** Borrowers or investors exercise options when interest rates change.

*Example:* A homeowner may repay a housing loan early when interest rates fall and refinance at a lower rate.

**How to manage interest rate risk**

- Match the maturity and repricing periods of assets and liabilities.
- Use fixed-rate or floating-rate borrowing according to the company's needs.
- Diversify investments across different maturities.
- Use interest rate futures, swaps, or options where appropriate.
- Conduct sensitivity analysis and stress testing.
- Maintain adequate cash reserves to manage increases in borrowing costs.

**Key point:** Interest rate risk cannot always be eliminated, but it can be measured, monitored, and reduced.

**2. Discounted Cash Flow (DCF)**

**Meaning**

**Discounted Cash Flow is a method of finding the present value of money expected to be received in the future.**

Money received today is generally worth more than the same amount received in the future because today's money can be invested to earn a return.

DCF helps investors and companies decide whether an investment is financially worthwhile.

**Formula**

$$PV = \frac{CF_t}{(1+r)^t}$$

Where:

- $PV$= Present value
- $CF_t$= Cash flow received in the future
- $r$= Discount rate per period
- $t$= Number of periods

The discount rate represents the required return or the rate used to adjust future cash flows for the time value of money and, depending on the analysis, risk.

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**Illustration**

Suppose a company expects to receive ₹11,000 after one year. Its required rate of return is 10%.

What is the present value?

$$PV = \frac{11{,}000}{1 + 0.10}$$

$$PV = \frac{11{,}000}{1.10} = ₹10{,}000$$

**Answer: ₹10,000.**

This means that receiving ₹10,000 today is financially equivalent to receiving ₹11,000 after one year, assuming a 10% annual discount rate.

**DCF for multiple cash flows**

Suppose an investment provides the following cash flows:

| **Year** | **Expected cash flow** |
|---|---|
| 1 | ₹5,000 |
| 2 | ₹6,000 |
| 3 | ₹7,000 |

If the discount rate is 10%, the present value of each cash flow is calculated separately.

$$PV = \frac{5{,}000}{1.10} + \frac{6{,}000}{1.10^2} + \frac{7{,}000}{1.10^3}$$

| **Year** | **Cash flow** | **Present value (approx.)** |
|---|---|---|
| 1 | ₹5,000 | ₹4,545.45 |
| 2 | ₹6,000 | ₹4,958.68 |
| 3 | ₹7,000 | ₹5,259.20 |

| | | |
|---|---|---|
| Total | ₹18,000 | ₹14,763.33 |

The investment's expected future receipts have a present value of approximately ₹14,763.

If the investment costs less than this amount, it may be attractive, provided the cash-flow estimates and discount rate are appropriate.

**Advantages of DCF**

- Considers the time value of money.
- Helps evaluate long-term investments.
- Accounts for cash flows occurring at different times.
- Supports capital budgeting and business valuation.
- Helps compare investment alternatives.

**Limitations of DCF**

- Future cash flows are estimates and may be inaccurate.
- Selecting the correct discount rate can be difficult.
- Results are sensitive to changes in assumptions.
- Unexpected market and economic conditions can affect actual returns.

**Remember:** A higher discount rate generally reduces the present value of a future cash flow.

**3. Time Value of Money (TVM)**

**Meaning**

**Time Value of Money means that money available today is generally worth more than the same amount of money received in the future.**

This is because money received today can be invested to earn interest.

For example, if you have ₹10,000 today and invest it at 10% annual interest, it will become ₹11,000 after one year. Therefore, ₹10,000 today and ₹10,000 received one year later are not financially equivalent.

**Important concepts**

**Present Value (PV):** The value of money today.

**Future Value (FV):** The value of money at a future date after earning interest.

**Interest rate:** The percentage earned or paid for using money.

**Compounding:** Calculating interest on both the original amount and previously earned interest.

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**Discounting:** Converting a future amount into its present value.

**Formula for Future Value**

$$FV = PV(1 + r)^n$$

Where:

- $FV$= Future value
- $PV$= Present value
- $r$= Annual interest rate
- $n$= Number of years

**Illustration**

Suppose ₹20,000 is deposited in a bank at 10% annual interest for two years, compounded annually.

$$FV = 20{,}000(1 + 0.10)^2$$

$$FV = 20{,}000(1.21)$$

$$\boxed{FV = ₹24{,}200}$$

The investment earns ₹4,200 over two years.

| Year | Opening amount | Interest at 10% | Closing amount |
|---|---|---|---|
| 1 | ₹20,000 | ₹2,000 | ₹22,000 |
| 2 | ₹22,000 | ₹2,200 | ₹24,200 |

Notice that second-year interest is calculated on ₹22,000, not just the original ₹20,000. This is the benefit of compounding.

**Formula for Present Value**

$$PV = \frac{FV}{(1 + r)^n}$$

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**Illustration**

You will receive ₹30,000 after three years. The discount rate is 10%.

$$PV = \frac{30{,}000}{(1.10)^3}$$

$$PV = \frac{30{,}000}{1.331}$$

$$\boxed{PV \approx ₹22{,}539.44}$$

Thus, ₹30,000 received after three years has a present value of approximately ₹22,539.44 at a 10% discount rate.

**Simple interest versus compound interest**

| **Basis** | **Simple interest** | **Compound interest** |
|---|---|---|
| Calculation | On the original principal | On principal plus accumulated interest |
| Interest earned | Usually lower over multiple periods | Usually higher over multiple periods |
| Example | Fixed interest on ₹10,000 | Interest grows as the balance increases |

For simple interest:

$$SI = \frac{P \times R \times T}{100}$$

For compound interest:

$$CI = P\left(1 + \frac{R}{100}\right)^T - P$$

**Applications of the time value of money**

- Calculating loan instalments.
- Evaluating business investments.

- Comparing deposits and investment schemes.
- Valuing bonds and shares.
- Planning retirement savings.
- Calculating the present value of future business income.

**Remember:** The higher the interest rate or the longer the waiting period, the lower the present value of a fixed future amount.

**4. Yield to Maturity (YTM)**

**Meaning**

**Yield to Maturity is the estimated annualised rate of return an investor earns by purchasing a bond at its current price and holding it until maturity, subject to certain assumptions.**

A bond is a loan made by an investor to a government or company. The issuer generally pays interest periodically and returns the principal at maturity.

YTM considers:

- The price paid for the bond.
- The interest received.
- The amount repaid at maturity.
- The time remaining until maturity.

YTM is a more complete return measure than the coupon rate because it also considers the difference between the purchase price and the repayment value.



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**Important terms**

**Face value:** The amount the issuer promises to repay at maturity.

**Coupon rate:** The stated annual interest rate on the bond's face value.

**Coupon payment:** The interest amount received by the investor.

**Market price:** The price at which the bond is currently bought or sold.

**Maturity:** The date on which the bond's principal is due for repayment.

**Illustration**

Suppose you buy a bond with the following features:

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| Particulars | Amount |
|---|---|
| Face value | ₹1,000 |
| Annual coupon rate | 8% |
| Annual coupon payment | ₹80 |
| Purchase price | ₹950 |
| Remaining maturity | 1 year |
| Repayment at maturity | ₹1,000 |

At the end of one year, you receive ₹80 as interest and ₹1,000 as principal.

Your total receipts are ₹1,080.

Your initial investment is ₹950.

$$\text{Return} = \frac{1{,}080 - 950}{950} \times 100$$

$$\boxed{\text{Return} = 13.68\%}$$

In this simplified one-year example, the yield is approximately 13.68%, assuming the issuer pays in full and there are no transaction costs or taxes.

Notice that the return is greater than the 8% coupon rate because the bond was purchased below its face value.

**Relationship between bond price and YTM**

| Bond price | Relationship | General result |
|---|---|---|
| Price equals face value | At par | YTM approximately equals coupon rate |
| Price is below face value | Discount | YTM generally exceeds coupon rate |
| Price is above face value | Premium | YTM generally falls below coupon rate |

These relationships assume a conventional fixed-coupon bond without unusual features.

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**Why does YTM change?**

Suppose a bond pays an annual coupon of ₹80 on a face value of ₹1,000.

If its market price falls from ₹1,000 to ₹900, an investor pays less to receive the same scheduled coupon and principal payments. Consequently, its YTM generally rises.

If the market price increases to ₹1,100, the investor pays more for those same payments, so its YTM generally falls.

**Easy rule to remember**

**Bond price falls**

Yield to maturity generally rises.

**Bond price rises**

Yield to maturity generally falls.

**Limitations of YTM**

- It is based on the assumption that scheduled payments are made in full.
- The realised return can differ if the bond is sold before maturity.
- The final return depends on the rates at which coupon payments are reinvested.
- Taxes, transaction costs, and default risk can affect actual returns.

**5. Fixed-Term Annuities and Present Value of Basis Points**

This part covers two important concepts used in financial risk management: annuities and the sensitivity of bond values to small interest-rate changes.

**5.1 Fixed-term annuity**

**Meaning**

**An annuity is a series of equal payments made or received at regular intervals for a specified period.**

For example, a person pays ₹10,000 every month towards a loan. A company receives ₹50,000 every year from a rental agreement. Both involve recurring cash flows.

An annuity may be:

- An ordinary annuity, where payments occur at the end of each period.
- An annuity due, where payments occur at the beginning of each period.

**Present value of an ordinary annuity**

The present value of an ordinary annuity is calculated using:

$$PV = C\left[\frac{1-(1+r)^{-n}}{r}\right]$$

Where:

- $C$= Payment per period
- $r$= Interest or discount rate per period
- $n$= Number of payments

**Illustration**

A person will receive ₹10,000 at the end of each year for three years. The discount rate is 10%.

Calculate the present value.

$$PV = 10{,}000\left[\frac{1-(1.10)^{-3}}{0.10}\right]$$

$$PV = 10{,}000(2.48685)$$

$$\boxed{PV \approx ₹24{,}868.52}$$

Although the total amount received over three years is ₹30,000, its present value is only about ₹24,868.52.

The difference arises because the payments are received at different times in the future.

**Applications of annuities**

- Home loan instalments.
- Vehicle loan repayments.
- Pension payments.
- Insurance premiums and benefits.
- Lease rentals.
- Fixed periodic investment plans.

**5.2 Present Value of a Basis Point (PVBP)**

**Meaning**

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**Present Value of a Basis Point, also called Price Value of a Basis Point, measures the approximate change in the value of a bond or fixed-income portfolio when its yield changes by one basis point.**

One basis point (bp) equals 0.01 percentage point.

For example:

- 5.00% to 5.01% = increase of 1 basis point.
- 5.00% to 5.10% = increase of 10 basis points.
- 5.00% to 6.00% = increase of 100 basis points.

PVBP is useful because even a small interest-rate change can produce a large monetary gain or loss when an investor holds a large bond portfolio.



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**Illustration**

Suppose a bond has a market value of ₹10,00,000 and a modified duration of 5 years.

An approximate formula is:

$$PVBP \approx P \times D_{\text{mod}} \times 0.0001$$

Where:

- $P$= Current market value.
- $D_{\text{mod}}$= Modified duration.
- $0.0001$= One basis point expressed as a decimal yield change.

Substituting:

$$PVBP \approx 10{,}00{,}000 \times 5 \times 0.0001$$

$$\boxed{PVBP \approx ₹500}$$

This means the bond's value is estimated to change by approximately ₹500 for a one-basis-point change in yield.

- If the yield rises by 1 bp, the bond loses approximately ₹500.
- If the yield falls by 1 bp, the bond gains approximately ₹500.

These are approximations for a small yield change, assuming other factors remain unchanged.

**Example of a larger change**

If the yield increases by 20 basis points:

$$\text{Estimated loss} = ₹500 \times 20$$

$$\boxed{\text{Estimated loss} \approx ₹10{,}000}$$

This is a duration-based estimate; the actual change may differ because bond prices do not respond in a perfectly linear manner.

**6. Measuring and Managing Market Risk**

**Meaning of market risk**

**Market risk is the possibility of financial loss caused by changes in market prices or market variables.**

These changes may affect shares, bonds, currencies, commodities, interest rates, and derivatives.

For example, an investor buys shares worth ₹1,00,000. If the share price falls by 10%, the investment's market value falls to ₹90,000. The investor faces a loss of ₹10,000 if the shares are sold at that price.

**Types of market risk**

**Equity price risk:** The risk that share prices will fall.

*Example:* A decline in the stock market reduces the value of an equity portfolio.

**Interest rate risk:** The risk that interest-rate movements will affect bond values or borrowing costs.

*Example:* Rising yields reduce the price of an existing fixed-rate bond.

**Foreign exchange risk:** The risk of loss caused by currency movements.

*Example:* An Indian importer must pay US$10,000. If the rupee weakens against the US dollar, the rupee cost of the payment increases.

**Commodity price risk:** The risk that the prices of raw materials or commodities will change.

*Example:* A jewellery manufacturer may face higher costs when gold prices rise.

**Volatility risk:** The risk associated with changes in the expected variability of market prices, particularly for options and other derivatives.

**Methods of measuring market risk**

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**Sensitivity analysis**

Sensitivity analysis estimates how much a financial position changes when one factor changes.

For example, if a share portfolio is worth ₹5,00,000 and falls by 2%, its value declines by:

$$5{,}00{,}000 \times \frac{2}{100} = ₹10{,}000$$

The estimated portfolio value becomes ₹4,90,000.

**Scenario analysis**

Scenario analysis examines what might happen under a particular market situation.

For example, a company may estimate its financial position under these scenarios:

- Share prices fall by 10%.
- Interest rates rise by 1 percentage point.
- The rupee depreciates by 5%.
- Gold prices rise by 8%.

The purpose is to understand how different market conditions affect profits and cash flows.

**Value at Risk (VaR)**

**Value at Risk is a statistical measure that estimates a potential loss over a specified period at a chosen confidence level, under a particular model and set of assumptions.**

For example, suppose a portfolio has a one-day VaR of ₹50,000 at 95% confidence.

This means the model estimates that, under its assumptions, the loss should not exceed ₹50,000 on approximately 95% of days. On roughly 5% of days, the loss could be greater.

VaR does not tell us the maximum possible loss. It also does not describe how severe losses may be beyond the VaR threshold.

**Stress testing**

Stress testing examines how a portfolio would perform under extreme but plausible conditions.

For example, a bank may test the effect of a severe stock-market decline, a sharp increase in interest rates, and a sudden withdrawal of deposits.

Stress testing helps identify weaknesses that ordinary risk measurements might miss.

**Methods of managing market risk**

- Diversify investments across assets and industries.
- Set limits on the size of individual positions.
- Use futures, options, and other derivatives for hedging.
- Monitor market prices and portfolio sensitivity.
- Maintain adequate liquidity.
- Conduct regular stress tests.

**Illustration:** An airline expecting to buy aviation fuel in the future may use appropriate commodity hedging instruments to reduce uncertainty about fuel costs.

The hedge may reduce losses if prices rise, but it may also reduce the benefit if prices fall. Hedging reduces selected risks rather than guaranteeing a profit.

**7. Risk–Return Trade-off**

**Meaning**

**The risk–return trade-off means that investments offering the possibility of higher returns generally involve higher levels of risk.**

Investors must decide how much risk they are willing and able to accept to achieve their financial goals.

For example, compare a bank deposit with an investment in shares. A deposit may offer a relatively predictable return, subject to the product's terms and applicable protections. Shares may provide higher returns, but their prices can rise or fall significantly.

**Illustration**

Consider three hypothetical investment choices.

| Investment | Expected annual return | Illustrative risk level |
|---|---|---|
| Bank deposit | 6% | Low |
| High-quality bond fund | 8% | Moderate |
| Equity investment | 14% | High |

These returns are examples for learning purposes, not current market offers or guaranteed results.

The equity investment has the highest expected return in this illustration, but its actual return could be negative.

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**How risk and return are related**

- Investors usually expect compensation for accepting additional risk.
- Higher expected returns do not guarantee higher actual returns.
- Riskier investments can produce both larger gains and larger losses.
- An investor's time horizon and ability to absorb losses matter.
- Diversification can reduce some risks without necessarily reducing expected returns by the same amount.

**Expected return**

When different outcomes are possible, expected return is calculated by multiplying each possible return by its probability and adding the results.

$$E(R) = \sum_{i=1}^{n} p_i \, R_i$$

Where:

- $E(R)$= Expected return
- $p_i$= Probability of an outcome
- $R_i$= Return under that outcome

**Numerical illustration**

An investment has two possible outcomes:

| Market condition | Probability | Return |
|---|---|---|
| Good market | 60% | 20% |
| Poor market | 40% | −10% |

Expected return:

$$E(R) = (0.60 \times 20\%) + (0.40 \times -10\%)$$

$$E(R) = 12\% - 4\%$$

$$\boxed{E(R) = 8\%}$$

The expected return is 8%. This does not mean the investor will definitely earn 8%; the actual return depends on which outcome occurs.

**Risk measurement using standard deviation**

Standard deviation measures how widely returns vary around their average.

A higher standard deviation generally indicates greater variability in returns. However, it does not distinguish between harmful downside movements and favourable upside movements.

For example, two investments may both have an average return of 10%, but one may fluctuate between 8% and 12%, while the other fluctuates between −20% and 40%. The second investment is generally much more volatile.

**Practical application**

A person saving for a near-term expense may prefer lower-risk investments because there is little time to recover from losses.

A person investing for a long-term goal may be able to accept some market fluctuations, depending on their circumstances.

**Key point:** The best investment is not necessarily the one with the highest expected return. It is the one whose risk is suitable for the investor's objectives and financial capacity.

**8. Systematic Risk and Idiosyncratic Risk**

Investment risk can be divided into two important categories: systematic risk and idiosyncratic risk.

**8.1 Systematic risk**

**Meaning**

**Systematic risk is the risk that affects the overall market or a large part of the economy. It cannot be eliminated simply by holding a large number of different shares.**

It is also called market risk in portfolio theory.

**Examples**

- Recession.
- High inflation.
- Major changes in interest rates.
- Global financial crises.
- Large-scale geopolitical conflicts.
- Broad market declines.

**Illustration**

Suppose an investor owns shares in 20 different Indian companies. If a major economic crisis causes the entire stock market to fall, the investor may suffer losses across many of those holdings.

Buying additional shares from different companies may not remove the effect of the economy-wide crisis.

### 8.2 Idiosyncratic risk

**Meaning**

**Idiosyncratic risk is the risk associated with a particular company, security, or organisation rather than the entire market.**

It is also known as company-specific or unsystematic risk.

**Examples**

- A company's poor financial results.
- A product failure.
- Fraud within an organisation.
- A strike affecting one company.
- The loss of a major customer.
- A company-specific regulatory problem.

**Illustration**

Suppose an automobile company experiences a product recall because of a manufacturing defect. Its share price may fall even if the wider stock market remains stable.

This is an example of company-specific risk.

**Comparison**

| **Basis** | **Systematic risk** | **Idiosyncratic risk** |
|---|---|---|
| Meaning | Risk affecting the broad market | Risk affecting a specific company or security |
| Main cause | Economic or market-wide events | Company-specific events |
| Example | Recession | Product recall |

| | | |
|---|---|---|
| Diversification | Cannot be eliminated through ordinary diversification | Can often be reduced through diversification |
| Common measure | Beta and market-factor sensitivity | Company-specific residual variability |

**How can investors manage these risks?**

**Systematic risk:** Investors can manage exposure through asset allocation, position sizing, suitable hedging strategies, and risk limits. It cannot be completely eliminated by simply diversifying within the same market.

**Idiosyncratic risk:** Investors can reduce exposure by spreading investments across different companies, industries, and other asset classes.

**Simple example**

An investor holds only one company's shares. If that company fails, the investor may lose a substantial part of the investment.

If the investor instead holds shares in many unrelated companies, a problem in one company may have a smaller effect on the overall portfolio.

However, diversification does not guarantee that the portfolio will avoid losses during a market-wide decline.

**9. Modern Portfolio Theory (MPT)**

**Meaning**

**Modern Portfolio Theory is an investment framework that explains how investors can combine different assets to balance risk and expected return.**

It is closely associated with economist Harry Markowitz.

The central idea is that investors should evaluate an investment not only by itself, but also by how it behaves in combination with the other investments in their portfolio.

For example, a portfolio containing only technology shares may be highly exposed to one industry. A portfolio containing technology shares, healthcare shares, bonds, and other assets may have a different overall risk profile.

**Important concepts**

**Portfolio**

A portfolio is a collection of investments held by an individual or organisation.

It may include shares, bonds, deposits, mutual funds, commodities, and other financial assets.

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**Diversification**

Diversification means spreading investments across different assets rather than concentrating all the money in one investment.

**Correlation**

Correlation measures how two investments move in relation to each other.

Correlation ranges from −1 to +1.

| Correlation | Meaning |
|---|---|
| +1 | The investments move perfectly together |
| 0 | No linear relationship in their movements |
| −1 | The investments move perfectly in opposite directions |

Lower or negative correlation can help reduce portfolio risk, although the degree of benefit depends on the assets' individual risks and portfolio weights.

**Expected portfolio return**

The expected return of a portfolio is the weighted average of the expected returns of its investments.

$$E(R_p) = \sum_{i=1}^{n} w_i \, E(R_i)$$

Where:

- $E(R_p)$= Expected portfolio return
- $w_i$= Proportion invested in asset $i$
- $E(R_i)$= Expected return of asset $i$

**Numerical illustration**

An investor has ₹1,00,000 and invests in two assets.

| Asset | Investment | Expected return |
|---|---|---|
| Shares | ₹60,000 | 12% |

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| | | |
|---|---|---|
| Bonds | ₹40,000 | 6% |

Expected portfolio return:

$$E(R_p) = (0.60 \times 12\%) + (0.40 \times 6\%)$$

$$E(R_p) = 7.2\% + 2.4\%$$

$$\boxed{E(R_p) = 9.6\%}$$

The expected annual return is 9.6%.

In rupee terms:

$$₹1{,}00{,}000 \times 9.6\% = ₹9{,}600$$

The expected return is ₹9,600 for the year, assuming the stated returns and weights apply. The actual return may differ.

**Portfolio risk**

Portfolio risk depends on:

- The risk of each investment.
- The proportion invested in each asset.
- The relationship between the assets' returns.
- Changes in market conditions.

For a two-asset portfolio, the variance of returns is:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}$$

Where:

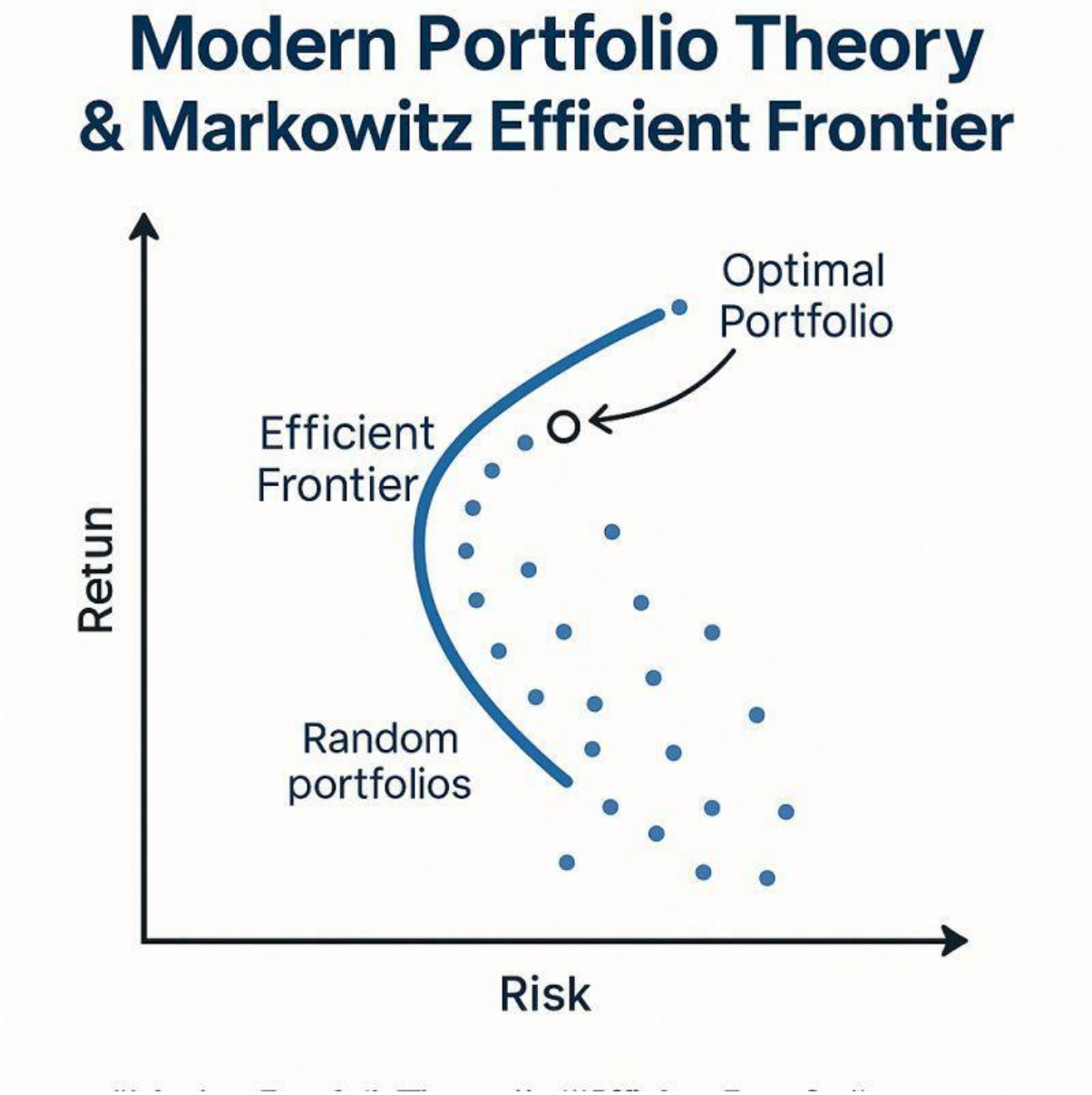
- $\sigma_p^2$= Portfolio variance
- $w_A, w_B$= Portfolio weights
- $\sigma_A, \sigma_B$= Standard deviations of asset returns
- $\rho_{AB}$= Correlation between the returns of the two assets

The correlation term is important. If two investments do not move together, combining them may reduce overall portfolio risk.

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**Efficient frontier**

The efficient frontier is the set of portfolios that offer the highest expected return for each level of risk, or the lowest risk for each target expected return, under the model's assumptions.




A portfolio below the efficient frontier is considered inefficient because another portfolio may provide a higher expected return for the same risk or lower risk for the same expected return.

**Advantages of Modern Portfolio Theory**

- Encourages diversification.
- Helps compare risk and expected return together.
- Supports rational portfolio construction.

- Helps identify inefficient portfolios.
- Provides a foundation for professional investment management.

**Limitations**

- Expected returns, volatility, and correlations must be estimated.
- Correlations may change during financial crises.
- Past behaviour may not predict future returns.
- Real-world investors may not behave as the model assumes.
- Transaction costs, taxes, and liquidity constraints can affect portfolio decisions.

**Remember:** Diversification is not simply holding many investments. It is holding investments whose risks and return patterns complement each other.

## 10. Return on Capital and Risk-Adjusted Return on Capital (RAROC)

### 10.1 Return on Capital

**Meaning**

Return on Capital measures how efficiently a company uses its capital to generate earnings.

A simple version is:

$$ROC = \frac{\text{Operating Profit}}{\text{Capital Employed}} \times 100$$

The exact formula depends on the purpose of the analysis. For example, analysts may use operating profit after tax rather than operating profit before tax.

**Illustration**

A company earns operating profit of ₹15 lakh and uses capital of ₹1 crore.

$$ROC = \frac{15{,}00{,}000}{1{,}00{,}00{,}000} \times 100$$

$$\boxed{ROC = 15\%}$$

The company generates operating profit equal to 15% of its capital employed.

### 10.2 Risk-Adjusted Return on Capital (RAROC)

**Meaning**

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**RAROC is a financial performance measure that evaluates the return earned in relation to the risk associated with the capital or business activity.**

It is particularly useful in banks and financial institutions, where two loans may generate similar income but have very different probabilities and potential costs of default.

A commonly used conceptual formula is:

$$RAROC = \frac{\text{Risk-adjusted return}}{\text{Economic capital}} \times 100$$

The exact definition of risk-adjusted return varies by institution. It may incorporate expected losses, operating costs, taxes, and other adjustments. Economic capital represents the capital allocated to absorb unexpected losses at a specified risk level.

**Illustration**

A bank considers two business loans.

| **Particulars** | **Loan A** | **Loan B** |
|---|---|---|
| Annual income | ₹12 lakh | ₹15 lakh |
| Expected loss and other risk adjustments | ₹2 lakh | ₹6 lakh |
| Risk-adjusted return | ₹10 lakh | ₹9 lakh |
| Economic capital | ₹50 lakh | ₹30 lakh |

For Loan A:

$$RAROC_A = \frac{10}{50} \times 100 = 20\%$$

For Loan B:

$$RAROC_B = \frac{9}{30} \times 100 = 30\%$$

Although Loan B generates a lower risk-adjusted return in rupees, it generates a higher return relative to the allocated economic capital.

**Loan A RAROC = 20%.**

**Loan B RAROC = 30%.**

Based on these illustrative figures alone, Loan B has the higher RAROC. A bank would still need to consider credit quality, concentration limits, liquidity, strategic priorities, and the reliability of the risk estimates.

**Why is RAROC important?**

- Helps compare loans with different risk levels.
- Supports risk-based pricing.
- Helps allocate capital efficiently.
- Assists banks in evaluating business units.
- Discourages decisions based only on gross revenue.
- Helps management compare returns with its required risk-adjusted hurdle rate.

**RAROC versus ordinary return on capital**

| **Basis** | **Return on Capital** | **RAROC** |
|---|---|---|
| Main focus | Profit relative to capital | Risk-adjusted return relative to capital |
| Risk adjustment | Not necessarily included | Explicitly incorporated |
| Common use | General corporate performance | Banking and financial risk management |
| Purpose | Measure capital efficiency | Assess whether returns justify the risk taken |

**11. Measuring and Managing Liquidity Risk**

**Meaning of liquidity risk**

**Liquidity risk is the possibility that a person, company, or financial institution will not have enough available cash to meet its financial obligations when they become due.**

A business may own valuable buildings, machinery, or inventory but still face a liquidity problem if it cannot convert those assets into cash quickly enough to pay salaries, suppliers, loan instalments, or other expenses.

In simple words, liquidity risk means **not having cash available at the right time**.

**Illustration**

Suppose a company has the following financial position.

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| Particulars | Amount |
|---|---|
| Cash available | ₹50,000 |
| Customer payments due next week | ₹2,00,000 |
| Supplier payment due tomorrow | ₹1,00,000 |
| Salary payment due in three days | ₹75,000 |

The company's immediate payments total ₹1,75,000, but it has only ₹50,000 in cash.

Even though customers owe it ₹2,00,000, the company may face difficulty if the customer payments arrive after the salaries and supplier payments are due.

This is a liquidity problem.

**Types of liquidity risk**

**Funding liquidity risk**

Funding liquidity risk occurs when an organisation cannot obtain enough cash to meet its payment obligations.

*Example:* A company cannot renew its short-term bank loan and does not have enough cash to repay it.

**Market liquidity risk**

Market liquidity risk occurs when an asset cannot be sold quickly at a reasonable price.

*Example:* An investor wants to sell a large quantity of shares in a thinly traded company. There may not be enough buyers at the expected price, forcing the investor to accept a lower price.

**Comparison**

| Funding liquidity risk | Market liquidity risk |
|---|---|
| Difficulty obtaining cash to make payments | Difficulty selling an asset at a fair price |
| Example: inability to repay a loan | Example: selling an illiquid bond at a discount |

| | |
|---|---|
| Managed through cash planning and funding arrangements | Managed through liquid investments and market-exposure limits |

**Causes of liquidity risk**

- Poor cash-flow forecasting.
- Delayed customer payments.
- Excessive short-term borrowing.
- Sudden withdrawal of bank deposits.
- Unexpected operating expenses.
- Difficulty selling investments.
- Reduced access to credit.
- Economic crises and market disruptions.

**Methods of measuring liquidity risk**

- **Current ratio**

The current ratio compares current assets with current liabilities.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Current assets generally include cash, receivables, and inventory. Current liabilities are obligations normally payable within the operating cycle or one year.

**Illustration**

Current assets = ₹6,00,000

Current liabilities = ₹3,00,000

$$\text{Current Ratio} = \frac{6{,}00{,}000}{3{,}00{,}000}$$

$$\boxed{\text{Current Ratio} = 2:1}$$

The company has ₹2 in current assets for every ₹1 in current liabilities.

A higher current ratio can indicate a stronger short-term position, but it does not automatically guarantee adequate liquidity. Inventory may be difficult to sell, and receivables may be collected late.

- **Quick ratio**

The quick ratio measures the ability to meet short-term obligations using assets that are relatively easy to convert into cash.

A common formula is:

$$\text{Quick Ratio} = \frac{\text{Current Assets} - \text{Inventory} - \text{Prepayments}}{\text{Current Liabilities}}$$

Suppose:

- Current assets = ₹6,00,000
- Inventory = ₹2,00,000
- Prepayments = ₹50,000
- Current liabilities = ₹3,00,000

$$\text{Quick Ratio} = \frac{6{,}00{,}000 - 2{,}00{,}000 - 50{,}000}{3{,}00{,}000}$$

$$\boxed{\text{Quick Ratio} \approx 1.17:1}$$

The company has approximately ₹1.17 in quick assets for every ₹1 of current liabilities.

- **Cash Flow Analysis** Cash-flow forecasting estimates when money will enter and leave the business. It helps identify shortages before they occur.

For example, a company can forecast cash balances for the next 30 days and identify a possible shortage before a large loan repayment falls due.

**Methods of managing liquidity risk**

- Maintain an adequate cash reserve.
- Prepare daily, weekly, and monthly cash-flow forecasts.
- Collect customer payments promptly.
- Negotiate suitable payment terms with suppliers.
- Avoid excessive dependence on short-term borrowing.
- Maintain access to appropriate credit facilities.

- Keep some investments in readily marketable assets.
- Establish contingency funding plans.
- Conduct stress testing for sudden cash outflows.

**Forecasting Cash Requirements from Cash Flow Analysis**

**Meaning of cash-flow analysis**

**Cash-flow analysis is the process of examining the cash entering and leaving a business during a particular period.**

It helps management understand whether the company has enough money to operate, pay its bills, repay loans, and invest in future growth.

Cash flow is different from profit. A company may record a sale as revenue but receive the cash from the customer several weeks later.

**Types of cash flows**

**Operating cash flow**

Cash generated or used by the company's normal business activities.

Examples:

- Cash received from customers.
- Payments to suppliers.
- Salary payments.
- Rent and other operating expenses.

**Investing cash flow**

Cash associated with purchasing or selling long-term assets and investments.

Examples:

- Purchasing machinery.
- Buying land.
- Selling equipment.
- Acquiring long-term investments.

**Financing cash flow**

Cash associated with raising or repaying capital.

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Examples:

- Taking a bank loan.
- Repaying loan principal.
- Issuing shares.
- Paying dividends.

**Cash-flow forecast**

A cash-flow forecast estimates future cash receipts and payments.

The basic formula is:

$$\begin{aligned}\text{Closing Cash} = &\text{ Opening Cash}\\ &+\text{Cash Inflows}\\ &-\text{Cash Outflows}\end{aligned}$$

**Numerical illustration**

A company wants to prepare a cash-flow forecast for January.

| **Particulars** | **Amount** |
|---|---|
| Opening cash balance | ₹1,00,000 |
| Cash received from customers | ₹4,00,000 |
| Other cash receipts | ₹50,000 |
| Payments to suppliers | ₹2,00,000 |
| Salaries | ₹1,00,000 |
| Rent and utilities | ₹30,000 |
| Loan repayment | ₹50,000 |

Total cash inflows:

$$₹4,00,000 + ₹50,000 = ₹4,50,000$$

Total cash outflows:

$$₹2,00,000 + ₹1,00,000 + ₹30,000 + ₹50,000 = ₹3,80,000$$

Closing cash balance:

$$₹1,00,000 + ₹4,50,000 - ₹3,80,000$$

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$$\boxed{\text{Closing Cash} = ₹1,70,000}$$

The company is expected to finish January with ₹1,70,000 in cash, assuming all estimated receipts and payments occur as planned.

**Forecasting a cash shortage**

Suppose the company expects the following position next month:

| **Particulars** | **Amount** |
|---|---|
| Opening cash | ₹1,70,000 |
| Expected cash receipts | ₹2,00,000 |
| Expected cash payments | ₹4,50,000 |

Closing cash:

$$₹1,70,000 + ₹2,00,000 - ₹4,50,000$$

$$\boxed{\text{Closing Cash} = -₹1,80,000}$$

The negative balance indicates a projected funding gap of ₹1,80,000.

Management should take action before the shortage occurs.

Possible actions include:

- Collecting receivables earlier.
- Delaying non-essential expenditure.
- Negotiating extended supplier credit.
- Arranging an appropriate bank overdraft or credit facility.
- Rescheduling payments where possible.
- Selling non-essential assets if appropriate.

The company should not simply assume that additional borrowing will always be available.

**Importance of cash-flow forecasting**

- Prevents unexpected cash shortages.

- Helps schedule loan repayments.
- Supports salary and supplier payments.
- Identifies the need for additional finance.
- Helps plan investment and expansion.
- Improves working-capital management.
- Supports financial risk management.

**Working Capital Management**

**Working capital is the difference between a company's current assets and current liabilities.**

It represents the short-term financial resources available to support daily business operations.

Current assets may include cash, inventory, and customer receivables. Current liabilities may include amounts owed to suppliers, short-term borrowings, and other short-term obligations.

**Formula**

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

**Illustration**

A company has the following balances:

| **Particulars** | **Amount** |
|---|---|
| Cash | ₹1,00,000 |
| Inventory | ₹2,00,000 |
| Receivables | ₹1,50,000 |
| Total current assets | ₹4,50,000 |
| Supplier payables | ₹1,00,000 |
| Short-term loan | ₹50,000 |
| Total current liabilities | ₹1,50,000 |

Working capital:

$$₹4{,}50{,}000 - ₹1{,}50{,}000$$

$$\boxed{\text{Working Capital} = ₹3{,}00{,}000}$$

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The company has positive working capital of ₹3,00,000.

This suggests that its current assets exceed its current liabilities. However, the company must still examine whether the inventory can be sold and receivables collected on time.

**Types of working capital**

- **Gross working capital**

Gross working capital refers to the total current assets of a company.

- **Net working capital**

Net working capital is current assets minus current liabilities.

- **Permanent working capital**

Permanent working capital is the minimum level of current assets required to keep the business operating continuously.

For example, a supermarket needs some inventory and cash every day, even when sales are low.

- **Temporary working capital**

Temporary working capital is the additional amount required to meet seasonal or unexpected increases in business activity.

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## Components of working capital

### Cash

Cash is used to pay salaries, suppliers, rent, electricity bills, and other immediate expenses.

### Inventory

Inventory includes raw materials, work in progress, and finished goods held for production or sale.

### Receivables

Receivables are amounts customers owe the company for goods or services already supplied.

### Payables

Payables are amounts the company owes its suppliers. They are current liabilities rather than current assets.

## Objectives of working capital management

- Maintain sufficient cash for day-to-day operations.
- Ensure that inventory is available when required.
- Collect money from customers on time.
- Pay suppliers according to agreed terms.
- Reduce unnecessary borrowing costs.
- Avoid excessive cash being tied up in inventory or receivables.
- Balance profitability with liquidity.

## Operating cycle

The operating cycle is the time taken to convert resources invested in inventory and production into cash collected from customers.

A simplified formula is:

$$\begin{aligned}\text{Operating Cycle} = {} & \text{Inventory Days} \\ & + \text{Receivables Days}\end{aligned}$$

A more complete calculation may include raw-material storage, work-in-progress time, finished-goods storage, and other stages of production.

## Illustration

Suppose a company takes:

- 40 days to sell its inventory.

- 30 days to collect payment from customers.

Its simplified operating cycle is:

$$40 + 30 = \boxed{70 \text{ days}}$$

If suppliers allow 20 days of credit, the simplified cash conversion cycle is:

$$\begin{aligned}\text{Cash Conversion Cycle} = {} & \text{Inventory Days} \\ & + \text{Receivables Days} \\ & - \text{Payables Days}\end{aligned}$$

$$40 + 30 - 20 = \boxed{50 \text{ days}}$$

The company must finance approximately 50 days of operating activity using its own funds or other financing, under these simplified assumptions.

A shorter cash conversion cycle can improve liquidity, provided it does not harm sales, supplier relationships, or operational efficiency.

**Methods of improving working capital**

**Inventory management:** Avoid excess stock while ensuring that sufficient goods are available to meet demand.

**Receivables management:** Check customer creditworthiness, issue invoices promptly, and follow up on overdue payments.

**Payables management:** Make use of agreed supplier credit terms without paying late or damaging business relationships.

**Cash management:** Maintain an appropriate cash balance and forecast upcoming receipts and payments.

**Short-term financing:** Use suitable working-capital facilities when necessary and manage borrowing costs carefully.

**Advantages of effective working capital management**

- Reduces the risk of cash shortages.
- Supports uninterrupted production.
- Helps pay employees and suppliers on time.
- Reduces unnecessary financing costs.
- Improves operational efficiency.

- Supports business growth and profitability.

**Problems caused by poor working capital management**

| **Problem** | **Possible consequence** |
|---|---|
| Excessive inventory | Money is tied up in unsold goods |
| Slow customer collections | Cash shortages and higher borrowing needs |
| Insufficient inventory | Lost sales and production delays |
| Excessive short-term borrowing | Higher interest costs and repayment pressure |
| Too little cash | Difficulty paying immediate obligations |
| Excessive idle cash | Funds may be underutilised |